Formula Sheet for Stat 332 Final Exam (Winter 2026)

Computational formulas for one-way ANOVA

$$\text{SSTO} = \sum_{i=1}^{I}\sum_{j=1}^{n_i} y_{ij}^2 - n\,\bar{y}_{\bullet\bullet}^2 \qquad \text{SSTR} = \sum_{i=1}^{I} n_i\,\bar{y}_{i\bullet}^2 - n\,\bar{y}_{\bullet\bullet}^2$$

$$\text{SSE} = \sum_{i=1}^{I}\sum_{j=1}^{n_i} y_{ij}^2 - \sum_{i=1}^{I} n_i\,\bar{y}_{i\bullet}^2$$

where

$$\bar{y}_{i\bullet} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij} \quad \text{and} \quad \bar{y}_{\bullet\bullet} = \frac{1}{n}\sum_{i=1}^{I}\sum_{j=1}^{n_i} y_{ij}$$

Computational formulas for two-way ANOVA

$$\text{SSTO} = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{n_0} y_{ijk}^2 - n_0\,I\,J\,\bar{y}_{\bullet\bullet\bullet}^2$$

$$\text{SSA} = n_0\,J\sum_{i=1}^{I}\bar{y}_{i\bullet\bullet}^2 - n_0\,I\,J\,\bar{y}_{\bullet\bullet\bullet}^2$$

$$\text{SSB} = n_0\,I\sum_{j=1}^{J}\bar{y}_{\bullet j\bullet}^2 - n_0\,I\,J\,\bar{y}_{\bullet\bullet\bullet}^2$$

$$\text{SSE} = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{n_0} y_{ijk}^2 - n_0\sum_{i=1}^{I}\sum_{j=1}^{J}\bar{y}_{ij\bullet}^2$$

$$\text{SSAB} = \text{SSTO} - \text{SSA} - \text{SSB} - \text{SSE}$$

where

$$\bar{y}_{i\bullet\bullet} = \frac{1}{n_0\,J}\sum_{j=1}^{J}\sum_{k=1}^{n_0} y_{ijk} \quad , \quad \bar{y}_{\bullet j\bullet} = \frac{1}{n_0\,I}\sum_{i=1}^{I}\sum_{k=1}^{n_0} y_{ijk}$$

$$\bar{y}_{ij\bullet} = \frac{1}{n_0}\sum_{k=1}^{n_0} y_{ijk} \quad \text{and} \quad \bar{y}_{\bullet\bullet\bullet} = \frac{1}{n_0 IJ}\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{n_0} y_{ijk} = \frac{1}{n}\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{n_0} y_{ijk}$$

Computational formulas for randomized block designs

$$\text{SSTO} = \sum_{i=1}^{I}\sum_{j=1}^{R} y_{ij}^2 - n\,\bar{y}_{\bullet\bullet}^2$$

$$\text{SSA} = R\sum_{i=1}^{I}\bar{y}_{i\bullet}^2 - n\,\bar{y}_{\bullet\bullet}^2$$

$$\text{SSBL} = I\sum_{j=1}^{R}\bar{y}_{\bullet j}^2 - n\,\bar{y}_{\bullet\bullet}^2$$

$$\text{SSE}_{\text{bl}} = \text{SSTO} - \text{SSA} - \text{SSBL}$$

where

$$\bar{y}_{i\bullet} = \frac{1}{R}\sum_{j=1}^{R} y_{ij} \quad , \quad \bar{y}_{\bullet j} = \frac{1}{I}\sum_{i=1}^{I} y_{ij} \quad \text{and} \quad \bar{y}_{\bullet\bullet} = \frac{1}{IR}\sum_{i=1}^{I}\sum_{j=1}^{R} y_{ij} = \frac{1}{n}\sum_{i=1}^{I}\sum_{j=1}^{R} y_{ij}$$

Computational formulas for 2^3 designs

$$\text{SS[XYZ]} = \frac{n}{4}(\text{effect}_{[XYZ]})^2 = \frac{n}{4}\left(\frac{1}{4}\sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2} c_{ijk}\,\bar{y}_{ijk\bullet}\right)^2$$

where [XYZ] is A, B, C, AB, etc.

$$\text{SSTO} = \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2}\sum_{\ell=1}^{r} y_{ijk\ell}^2 - 8\,r\,\bar{y}_{\bullet\bullet\bullet\bullet}^2$$

$$\text{SSE} = \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2}\sum_{\ell=1}^{r} y_{ijk\ell}^2 - r\sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2}\bar{y}_{ijk\bullet}^2$$

where

$$\bar{y}_{ijk\bullet} = \frac{1}{r}\sum_{\ell=1}^{r} y_{ijk\ell} \quad \text{and} \quad \bar{y}_{\bullet\bullet\bullet\bullet} = \frac{1}{n}\sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2}\sum_{\ell=1}^{r} y_{ijk\ell}$$

Least squares estimators

Least squares estimators are obtained by minimizing the following equation

$$Q = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{n_0}(y_{ijk} - \hat{y}_{ijk})^2 = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{n_0}\left(y_{ijk} - (\hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j + (\widehat{\alpha\beta})_{ij})\right)^2$$

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F Distribution Table

d.f.D.	α = 0.05																		
	d.f.N.																		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**12**	**15**	**20**	**24**	**30**	**40**	**60**	**120**	**∞**
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.52	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

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